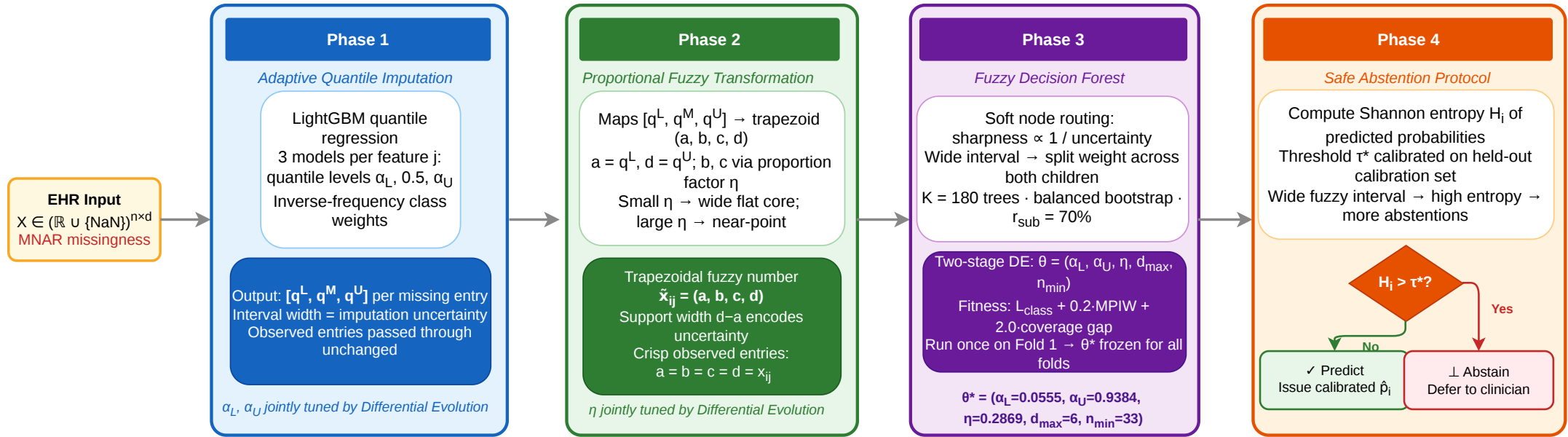


JUCO: Four-Phase Pipeline — Missing Data → Interval → Fuzzy → Soft Routing → Entropy → Abstention



DE jointly optimises all 5 parameters: α_L, α_U (Phase 1) $\cdot \eta$ (Phase 2) $\cdot d_{\text{max}}, n_{\text{min}}$ (Phase 3). Optimisation runs once on Fold 1 of MIMIC-IV; θ^* is frozen for Folds 2–5 and eICU without modification.